Pooled testing for HIV viral load: simulation of reaction use and time to confirmation of failure, with parameters from the Brazilian public network

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Abstract. Background: in HIV treatment programs with high viral suppression, most routine viral load tests fall below the failure cutoff. In Brazil, about 89% of results from people on treatment for six months or more are below 50 copies/mL, and the public network contracts around 1.1 million tests per year. Testing samples in pools saves reactions, but dilution, low-level viremia and residual RNA below 50 copies/mL may delay the detection of failure at the 200 cutoff or consume the savings. This note records the technique and estimates its potential, using Brazilian parameters as a reference case. Methods: simulation with the national viral load distribution (2022), detectable residual RNA in 25% of collections below 50 copies/mL in the main scenario, drawn between 1 and 49 copies/mL (in the sensitivity analysis, shares from 0% to 40% and ranges from 20 to 49 and from 40 to 49) and the detection, quantification and reproducibility described in the instructions for use of the network kits (cobas HIV-1 on the 5800 and 4800 systems). Retest rules were restricted to those the kit allows to be executed: retest only pools quantified at 20 cp/mL or more (rule Q20, main) or any detected pool (rule D). In the cross-sectional stage (400,000 collections), strategies were compared by subrange, with discordances in both directions. In the longitudinal stage, a cohort of 200,000 suppressed adults was followed for 24 months, with collections every six months, confirmation in a new collection 28 days after the report and the same 17,301 episodes in both arms. Variation comes from 40 replicates. Results: pools of 4 with rule Q20 used 39.7% fewer reactions per collection (empirical interval across replicates 39.5% to 40.0%) and 50.4% fewer during follow-up (50.3% to 50.5%), with a simulated difference close to zero in confirmation within 60 days (0.0 point; -0.3 to 0.2) and a median of 47 days until confirmation, against 42 for individual testing. In the residual RNA scenarios tested, rule Q20 saved 26% to 42% of reactions per collection and 35% to 53% during follow-up; rule D, from 38% to nothing. Just above the cutoff, confirmation within 60 days had wide intervals (rebounds of 200 to 249: -1.0 point; -6.7 to 4.2) and fell by about 3 points between 200 and 299 with greater imprecision. Quantitative deconvolution saved 56% of reactions, but detected only 35.0% of loads between 200 and 249, against 65.5% for individual testing. Pre-concentration of plasma showed no advantage: with rule D, it saved 6% to 7%; with a trigger equivalent to that of Q20, it tied with the ordinary pool. Conclusion: in this simulation, retesting only quantified pools kept savings in all the scenarios tested, from 26% to 53% of reactions, with a magnitude sensitive to the distribution of loads below 50, and confirmation of failure close to that of individual testing overall, with more uncertainty just above the cutoff. The cost reduction compared with current practice would be a fraction of these percentages, proportional to the eligible fraction and to the weight of the reaction in the price, and would depend on laboratory validation, on admission of the procedure and on payment linked to the number of reactions.

Keywords: HIV viral load; pooled testing; simulation; virological failure; residual RNA; test costs.

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https://mais-saude-ai.vercel.app/en/notes/testagem-grupo-carga-viral-hiv/

Translated from the Portuguese original.

In plain language

Why did we do this study? People who live with HIV and take treatment repeat a blood test a few times a year to see if the virus is under control. Almost always the result says there is no virus. We wanted to know if it is possible to use fewer tests without taking longer to find out when the treatment stops working.

What did we do and what did we find? We used a computer to try out an old idea: mix a little of the blood of four people and test the mix only once. If the mix shows little or no virus, no one in the group needs a new test. If it shows real virus, each person is tested alone. We used the numbers of the test used in Brazil. Many people on treatment still have a tiny bit of virus left in the blood. So the rule that worked only repeats the test when the mix has enough virus to be counted. This way, the idea would use between a quarter and a half fewer tests. Almost all the problem cases would be confirmed in the same time, with about five more days of waiting. When the virus came back only a little, the group may let a few more cases slip through.

What does this mean? The idea works well on the computer, but it still needs to be tried in a lab. To turn into money saved, rules and the way the tests are paid for would also need to change, and only part of the tests could use the idea. The note keeps the numbers on record for when this is possible. It says nothing about the health of any person.

Revision notes

Version 1.2 (Nov 9, 2026). A second review showed that the savings of rule Q20 also depend on where the residual RNA below 50 is concentrated, and not only on how many collections have it; that the aggregate 60-day result hid a larger loss just above the cutoff; and that pre-concentration had been tested only with rule D. Version 1.2 adds scenarios with residual RNA between 20 and 49 and between 40 and 49 cp/mL, confirmation within 60 days by rebound band and pre-concentration with a trigger equivalent to that of rule Q20, and now presents the savings as a range. Version 1.1 remains accessible at commit 6bcaa43.

Version 1.1 (Nov 9, 2026). Version 1 assumed that all collections below 50 cp/mL had no RNA at all. A review of the code showed that this premise underpinned the savings of rule D, which retests any detected pool. The literature indicates that a relevant share of suppressed people have detectable residual RNA below 20 cp/mL. Version 1.1 includes this residual RNA (25% of collections below 50 in the main scenario, with sensitivity from 0% to 40%), replaces the main rule with rule Q20, which retests only quantified pools and is barely affected by it, uses the same episodes in both arms of the follow-up, renames the secondary outcome to confirmation within 240 days and now shows the economic estimate as a gross scenario, to be multiplied by the eligible fraction. Version 1 remains accessible in the public repository, at commit 10f9dcd.

I. INTRODUCTION

A. The problem and why it matters

The viral load test is the main way to know whether HIV treatment is working. The more effective the treatment, the larger the share of routine tests with no quantifiable virus. In Brazil, in 2022, 89% of people on treatment for six months or more had fewer than 50 copies/mL (Table II), and the public network works with about 1.1 million HIV viral load tests per year, at a reference price of R$ 67.31 per test in 2026 [15].

When most samples are below the cutoff, testing them in pools can save reactions. The technical question is whether this is possible without delaying the identification of those in failure, which, under the operational rule used in this note, means two consecutive results of 200 copies/mL or more, with at least 4 weeks between collections [12].

B. What is already known

Pooled testing was proposed in the 1940s [1] and has been used for decades in blood banks, including with plasma concentration by centrifugation to offset dilution [2].

For HIV viral load during treatment, studies from 2009 to 2017 evaluated pools of 5 and 10 $\times$ 10 matrices with quantitative deconvolution, with reductions from one third to more than half in the number of tests [3–6]. There are also prediction-guided methods [7]. In 2025, a reference laboratory in Cameroon reported routine use of pools of 3, with 0.55 assays per result [8].

Below 50 copies/mL, “suppressed” does not mean free of RNA. Current kits distinguish “não detectado” from “detectado, inferior a 20”. In a Spanish service, 25.3% of people on suppressive treatment had between 20 and 49 copies/mL with an earlier kit [22]; a review reports very low-level viremia in 20% to 60% of people on suppressive treatment, depending on the kit [23]; and, in a French cohort, switching to the cobas 6800 system doubled the share of patients with some result of 20 copies/mL or more, from 18.5% to 37.0% [21]. This matters for pooled testing because a rule that retests any detection also ends up retesting these pools.

These works were found through a general internet search and by reading abstracts, on Nov 8 and 9, 2026. There was no systematic literature search: PubMed, SciELO, LILACS, Europe PMC and other databases were not searched systematically, and most of the full texts were not read. For this reason, this note does not claim that the approach is new. As far as the search reached, many earlier studies prioritized thresholds higher than 200 copies/mL and used kits that predate the current ones.

C. The question

This note asks, with Brazilian public parameters:

- **Performance per collection:** how many reactions does each pooling strategy save, and how does it compare with individual testing in detecting loads between 200 and 249, 250 and 299, 300 and 499 and 500 and 999 copies/mL?
- **Performance during follow-up:** among suppressed adults, does the best strategy confirm virological failure within 60 days in the same proportion as individual testing, and with how much delay?
- **Residual RNA:** how much do the savings of each rule depend on the share of collections with detectable RNA below 50 copies/mL?
- **First principles:** does concentrating the pool’s plasma, to avoid dilution, improve the result?
- **Potential:** if the conditions for adoption converged, how large would the cost reduction be compared with current practice?

D. What this note does and does not do

The note records the technique, simulates its performance and estimates its economic potential in a conditional way. The network’s procurement documents are used only as a reference for the scale, price and payment model of current practice. The note does not test real samples, does not assess laboratories, contracts or suppliers, does not guide the follow-up of any person and does not replace bench validation.

II. DATA

Below the lowest concentration with data (50 cp/mL on the 5800 and 77 on the 4800), the standard deviation was assumed to be 0.25 log10 at 20 cp/mL. The robustness scenario used 0.40 below 50 cp/mL, 0.20 at 200 and 0.12 from 1,000 upward.

III. METHODS

A. Design overview

There are two stages, with the same assay parameters. The cross-sectional stage compares strategies in a single round of collections. The longitudinal stage follows people over time and measures the time until confirmation. A third part converts the reduction in reactions into economic potential, in a conditional way. All code uses fixed seeds. The note has no protocol registered before the analyses, and all results are exploratory.

B. Measurement model

The chance that the kit detects the virus in a reaction with concentration c was modeled as

$$P(\text{detect}) = 1 - e^{-3c/L} \tag{1}$$

where L is the 95% limit of detection: with $c = L$, the chance is $1 - e^{-3} \approx 95\%$. The model reproduces the detection rates in the instructions for use at 20, 10 and 5 cp/mL with a difference of up to 2 points (Table III) and overestimates detection at 2.5 cp/mL (43% against 35%), which only sends some pools to unnecessary retest. The measured value follows a lognormal distribution around the true value, with the standard deviation in Table III.

C. Residual RNA below 50

In a fraction f of collections below 50 cp/mL, the true load is drawn uniformly on a log scale between 1 and 49 cp/mL; in the others, it is zero. With this shape, about 23% of collections with residual RNA fall between 20 and 49, and the rest below 20. The main scenario uses $f = 25\%$, in the order of magnitude of the studies cited [21–23]; the sensitivity analysis uses 0% (the premise of version 1), 10% and 40%. The shape matters as much as the share: with it, only 5.8% of collections below 50 fall between 20 and 49, while the Spanish study found 25.3% of people in that range, with one collection per person and a different kit [22]. The two quantities are not the same, but the study justifies scenarios with more residual RNA close to 50. Therefore, keeping the share at 25%, the note also draws residual RNA between 20 and 49 and, as a stress test, between 40 and 49 cp/mL. During follow-up, the draw is repeated at each collection, for those who are not in rebound or in transient viremia.

D. Reference

Individual testing. Each sample is tested once; the result is elevated when the measurement is 200 cp/mL or more.

E. Pooling strategies

The pool receives equal parts of k samples. The retest rules were restricted to those executable with the kit's output:

- **Rule Q20 (main):** the pool goes to retest only when it is quantified at 20 cp/mL or more. In a pool of 4, this is equivalent to 80 cp/mL per sample.
- **Rule D (comparison):** the pool goes to retest whenever the kit detects virus, including when the result is "detectado, inferior a 20 cp/mL", with no numerical value.

Numerical triggers below 20 cp/mL in the pool were not used, because the kit does not report a value in that range. The retest is a new measurement of the stored sample, independent of the measurement in the individual arm. Pools of 3, 4, 5 and 8 were evaluated with rule Q20, and of 3 and 4 with rule D. Two comparisons complete the set:

- **Quantitative deconvolution** with pools of 5 and a threshold of 250 cp/mL (configuration simulated from [3, 4]): if the pool indicates 250 or more per sample, the samples are tested one by one, and the value of each is subtracted from the estimated total until the remainder falls below the threshold. Only quantified values enter the calculations.
- **Pre-concentration:** 1 mL from each of 4 samples is concentrated into the volume of one reaction, with mean recovery of 0.7 and 0%, 2% or 5% of poor recoveries (between 0 and 0.2). It was tested with rule D and with a trigger of 56 cp/mL in the concentrate, equivalent to that of rule Q20 (20 cp/mL in the pool $\times$ 4 samples $\times$ recovery of 0.7).

Example

In a pool of 4 with one sample at 240 cp/mL and three with no virus, the pool has about 60 cp/mL: the kit quantifies it, and the pool goes to retest under both rules. If, instead of the sample at 240, the pool has one person with residual RNA of 15 cp/mL, the pool will have about 4 cp/mL: the kit may report "detectado, inferior a 20", and the pool goes to retest under rule D, but not under rule Q20.

F. Cross-sectional stage

400,000 collections were simulated with the distribution in Table II, drawn uniformly on a log scale within each range, with the residual RNA described above. For each strategy, reactions per sample and the share detected as 200 or more in each subrange were measured. Discordances with individual

Table I. Public sources used, with period and main limitation.

Source	What it provides	Period	Main limitation
HIV Clinical Monitoring Report 2022, Figure 76 [9]	national distribution of viral load by range	Jan to Sep 2022	includes everyone on treatment for 6 months or more, not only routine tests of suppressed people; does not separate "não detectado" from "detectado" below 50; values read from a chart
Instructions for use of cobas HIV-1 for cobas 5800/6800/8800 [10]	detection near the limit, limit of quantification, reproducibility, minimum volume, how results are reported	version consulted on Nov 8, 2026	precision measured on the 6800/8800 systems; no imprecision data below 50 cp/mL
Instructions for use of cobas HIV-1 for cobas 4800 [11]	the same items for the 4800	version consulted on Nov 8, 2026	no imprecision data below 77 cp/mL
Studies on residual RNA during suppressive treatment [21–23]	order of magnitude of the share of people with detectable RNA below 50 cp/mL	2010 to 2022	services in other countries, different kits and definitions; shares per person, not per collection
Technical Notes 286/2023 and 74/2026 [12, 13]	network kits and result categories; operational definition of confirmed failure	2023 to 2026	summary of the instructions for use; genotyping context
Network procurement documents [14–17]	annual scale, reference price, payment model and execution rules of current practice	2021 to 2026	contracted quantities, not performed

Table II. Viral load distribution used in the simulations: people aged 2 years or older, on treatment for at least six months, Brazil, January to September 2022 [9]. The range below 50 receives residual RNA in a share of collections (section III).

Range (copies/mL)	Proportion
below 50	89%
50 to 199	4%
200 to 999	2%
1,000 to 9,999	2%
10,000 or more	3%

testing were counted in both directions, with independent measurements in both arms. The loss caused by the retest rule was separated out: cases that individual testing detects and whose pool did not go to retest.

G. Longitudinal stage

A cohort of 200,000 suppressed adults was followed for 24 months, with routine collections at months 6, 12, 18 and 24.

- Sustained rebounds arise with an annual risk of 6%. Two in seven stay between 200 and 999 cp/mL, and the rest above 1,000, in the proportion of Table II.
- Transient viremias between 50 and 199 cp/mL appear in 4% of collections, as in Table II.
- Each elevated collection triggers a new collection 28 days after the report, always tested individually. Two results of 200 or more confirm failure, and the person leaves the pools.
- The report takes 7 days, plus 5 days when the pool goes to retest. This addition is fixed in the model and on its own explains the difference in median between the arms.

The operational rule follows Technical Note 74/2026 ($\geq$ 200 cp/mL); the 2024 PCDT uses the wording "$>$ 200", and the difference does not change the simulation.

Episodes: the same in both arms, defined by the cohort alone: the first routine collection after the start of the rebound, up to month 18 (17,301 episodes). **Primary outcome:** proportion of episodes confirmed within 60 days after that collection, counting in the denominator the episodes that the strategy missed. **Secondary outcomes:** confirmation within 240 days (the time to the next routine collection, six months later, plus 60 days) and median days until confirmation.

H. Economic potential

The reduction in reactions does not convert fully into a cost reduction, because the price per test also includes equipment, collection, transport, maintenance, technical support and training [16, 17], and because only part of the tests would be eligible. The potential was estimated as

$$\text{annual reduction} = R \times e \times w \times V \tag{2}$$

where R is the reduction in reactions among eligible tests, e is the eligible fraction of the volume, w is the weight of the reaction in the price per test and V is the annual reference value (1.1 million tests at R$ 67.31, about R$ 74 million [15]). As e and w are not public, the note shows the gross scenario with $e = 100\%$ and illustrative weights of 30%, 50% and 60%,

Table III. Assay parameters used in each scenario, from the instructions for use.

Parameter	cobas 5800 (main scenario)	cobas 4800 (sensitivity)
Processing volume	500 μL	400 μL
Minimum volume in the tube	650 μL	1,000 μL (secondary tube)
Limit of detection (95%)	13.2 cp/mL	14.2 cp/mL
Detection at 20, 10 and 5 cp/mL	98.4%, 90.5% and 66.1%	98.0%, 90.1% and 63.5%
Limit of quantification	20 cp/mL	20 cp/mL
Result below quantification	"HIV detetado, inferior a 20 cp/ml"	"HIV detectado, inferior a 20 cp/ml"
Standard deviation (log10) used	0.191 at 50; 0.116 at 200; 0.102 at 400; 0.092 at 1,000; 0.081 to 0.084 above	0.19 at 77; 0.09 at 308 and 770; 0.06 at 7,700; 0.05 above
Source of the standard deviation	reproducibility in 3 centers (6800)	within-laboratory precision

and an example with $e = 60\%$. The additional costs of the pooled workflow (handling, tracking, reporting systems) were not deducted.

I. Variation

The intervals are empirical across 40 replicates with different seeds (2.5% to 97.5%). They describe the variation under the simulated model, not the uncertainty of the parameters, which was explored through the scenarios and the sensitivity analysis. Numbers from different tables may vary in the decimal place because they come from different draws.

J. Sensitivity analyses

- Share of collections below 50 with residual RNA: 0%, 10%, 25% and 40%.
- cobas 4800 parameters instead of the 5800.
- Robustness scenario with greater imprecision (Table III).
- Annual rebound risk of 3%, 6% and 10%, crossed with transient viremia in 4% and 8% of collections.

K. Protocol deviations

Not applicable: the protocol has not yet been registered. The rules and configurations were revised during exploration, after earlier results had been examined, including through external reviews that pointed out a trigger that could not be executed with the kit (before version 1) and the premise of zero load below 50 (version 1.1). In a confirmatory version, the main rule, the share of residual RNA, the handling of non-quantifiable results and the 60-day outcome will be fixed before the run.

IV. RESULTS

How to read these results

- "Fewer reactions" compares with individual testing on the same collections.
- Detection is the share of samples in each subrange that received a result of 200 or more. Individual testing does not reach 100% near the cutoff because imprecision pushes part of the loads close to 200 below it.
- Differences of 1 to 3 points in a subrange are within the variation expected between two measurements of the same sample (Table V).
- Unless otherwise stated, the results use the main scenario: cobas 5800 and residual RNA in 25% of collections below 50.

A. Performance per collection

Across the 40 replicates, the pool of 4 with rule Q20 used 39.7% fewer reactions (empirical interval 39.5% to 40.0%), with a difference of -0.1 point (-1.3 to 0.8) in detection between 200 and 499 cp/mL. Pools of 5 saved almost the same, and pools of 8 saved less and detected less near the cutoff, because dilution brings more loads just above 200 below 20 in the pool. Rule D saved less than half as much as rule Q20, because residual RNA sends many pools to retest. Quantitative deconvolution saved more, but detected only 35.0% of loads between 200 and 249 and 83.9% between 300 and 499.

Discordances in both directions reflect the repeatability of the assay near the cutoff, which also appears between two individual measurements of the same sample. The loss caused by the retest rule was 2 in 1,106 samples between 200 and 249 cp/mL under rule Q20, and zero above 250; under rule D, it was zero.

Table IV. Cross-sectional stage, main scenario (cobas 5800, residual RNA of 25%): fewer reactions and detection by subrange (cp/mL). In bold, the main strategy.

Strategy	Fewer reactions	200 to 249	250 to 299	300 to 499	500 to 999	1,000 or more
Individual	0%	65.5%	89.1%	99.1%	100%	100%
Pool of 3, rule Q20	36.3%	67.5%	87.5%	99.0%	100%	100%
Pool of 4, rule Q20	**39.4%**	**67.5%**	**88.6%**	**99.3%**	**100%**	**100%**
Pool of 5, rule Q20	40.0%	65.1%	87.9%	98.8%	100%	100%
Pool of 8, rule Q20	36.5%	60.4%	81.2%	98.3%	100%	100%
Pool of 3, rule D	14.9%	69.4%	89.4%	99.2%	100%	100%
Pool of 4, rule D	16.0%	67.0%	88.3%	99.1%	100%	100%
Pool of 5, deconvolution (threshold 250)	56.3%	35.0%	62.5%	83.9%	95.6%	98.6%
Pre-concentration of 4, no failures	6.3%	66.8%	89.0%	99.2%	100%	100%
Pre-concentration of 4, 5% failures	7.4%	65.0%	88.7%	98.9%	99.9%	100%
Pre-concentration of 4, trigger 56, no failures	39.3%	66.6%	89.3%	98.8%	100%	100%
Pre-concentration of 4, trigger 56, 5% failures	40.0%	62.7%	83.9%	95.9%	98.7%	99.8%

Table V. Discordances between individual testing and the pool of 4, with independent measurements in both arms, main scenario. The last column is the loss caused by the retest rule.

Rule and subrange (cp/mL)	Samples	Only individual detects	Only pool detects	Individual detects and pool not retested
Q20, 200 to 249	1,106	249	219	2
Q20, 250 to 299	894	83	71	0
Q20, 300 to 499	2,567	16	13	0
Q20, 500 to 999	3,512	0	0	0
D, 200 to 249	1,106	247	220	0
D, 250 to 299	894	83	71	0
D, 300 to 499	2,567	16	13	0

B. Sensitivity to residual RNA

The savings of rule D fall from 38% to 6% per collection when the share of residual RNA goes from 0% to 40%. Those of rule Q20 fall only from 42% to 38%. One person's residual RNA, diluted in four, almost always stays below 20 cp/mL in the pool, and therefore does not trigger rule Q20. In all shares, the losses from lack of retest under rule Q20 stayed at 2 samples between 200 and 999 cp/mL, and confirmation within 60 days did not change noticeably.

When residual RNA is concentrated close to 50, the savings of rule Q20 fall to 32% and 26% per collection, because two residues in the same pool more often add up to 20 cp/mL or more in the pool. Rule D comes to use as many reactions as individual testing, or more. The savings of Q20 in the scenarios tested therefore lie between 26% and 42% per collection and between 35% and 53% during follow-up, depending on the share and the shape of the residual RNA.

C. Pre-concentration

Pre-concentration showed no advantage in the configurations evaluated. With rule D, concentrating 1 mL of each sample makes the residual RNA detectable almost always, and the savings fall to 6% to 7%. With the trigger equivalent to that of Q20, the savings rise to 39% to 40%, practically the same as the ordinary pool, before the extra cost of concentrating; and, with 5% of poor recoveries, detection falls between 300 and 999 cp/mL (95.9% and 98.7%). Dilution is not the bottleneck with the current kits: a pool of 4 takes a sample of 200 cp/mL to about 50 in the pool, which is quantified almost always.

D. 24-month follow-up

Across the 40 replicates, the pool of 4 with rule Q20 used 50.4% fewer reactions (empirical interval 50.3% to 50.5%), with a difference of 0.0 point (-0.3 to 0.2) in confirmation within 60 days. The median of 47 days corresponds to the direct path: report in 7 days plus 5 for retest, new collection 28 days later and confirmation report in 7 days. The savings during follow-up are larger than those per collection because the cohort only includes suppressed people, and those with confirmed failure leave the pools.

The aggregate result is dominated by high rebounds: 71% of episodes are at 1,000 cp/mL or more, and almost all are confirmed in both arms. Just above the cutoff, confirmation within 60 days is low even for individual testing, because the imprecision of the assay takes part of the measurements below 200, and the pool loses a little more (Table IX).

In the main scenario, the differences by band are compatible with zero, but with wide intervals between 200 and 299 (up to 6.7 points less). With greater imprecision, the pool confirmed about 3 points less between 200 and 299 and 1 point less between 300 and 499. The difference close to zero holds for the simulated set as a whole; just above the cutoff, there is

Table VI. Fewer reactions with pools of 4 under rules D and Q20, per collection and during follow-up, and confirmation within 60 days (individual, D and Q20), depending on the share of collections below 50 with residual RNA. Cobas 5800. The 0% row reproduces the premise of version 1.

Residual RNA	Per collection, D	Per collection, Q20	Follow-up, D	Follow-up, Q20	Within 60 days
0%	37.6%	41.6%	48.5%	52.9%	97.0%, 97.1% and 97.0%
10%	28.1%	41.0%	37.7%	52.1%	96.8%, 96.9% and 97.1%
25%	16.0%	**39.5%**	23.6%	**50.5%**	96.9%, 97.0% and 96.9%
40%	6.1%	37.7%	11.9%	48.3%	97.1%, 97.0% and 96.9%

Table VII. Range in which residual RNA is drawn, with 25% of collections below 50 having residual RNA: fewer reactions with pools of 4 (rules D and Q20, per collection; Q20, during follow-up) and confirmation within 60 days (individual and Q20). Cobas 5800.

Residual RNA range	Per collection, D	Per collection, Q20	Follow-up, Q20	Within 60 days
1 to 49 cp/mL	16.0%	**39.5%**	**50.5%**	96.9% and 96.9%
20 to 49 cp/mL	0.1%	32.1%	42.1%	96.8% and 96.9%
40 to 49 cp/mL (stress)	-2.8%	26.0%	35.2%	97.0% and 96.9%

more uncertainty and a probable loss when the assay is less precise.

E. Sensitivity

In the robustness scenario, rule Q20 detected 56.2% of loads between 200 and 249, against 60.1% for individual testing, and confirmed 0.2 point less within 60 days overall and about 3 points less between 200 and 299 (Table IX). This is the loss just above the cutoff that greater imprecision produces when the pool is close to 20 cp/mL.

In all combinations, the difference in confirmation within 60 days was 0.3 point or less, in either direction. The reduction in reactions ranged from 36% to 56%, mainly depending on the frequency of transient viremia.

F. Plasma volume

In the 500 μL workflow of the cobas 5800, the tube needs at least 650 μL [10]. A pool of 4 would require about 163 μL from each sample, and a possible individual retest another 650 μL. Each collection would need to yield at least about 0.8 mL of plasma, plus a reserve for repetition. On the cobas 4800, with 1,000 μL in the secondary tube, it would be about 1.25 mL [11].

G. Conditional economic potential

The table assumes that the whole volume would be eligible, which is not realistic: the technique would only apply to routine tests of people already suppressed. With 60% eligibility, for example, the range would fall to 7.1% to 18.2% of the annual value, still before the additional costs. And, if residual RNA is concentrated close to 50 cp/mL (26% fewer reactions per collection and 35% during follow-up), the gross scenario would range from 7.8% to 21.1%, and, with 60% eligibility, from 4.7% to 12.7%. And these values would only materialize if four conditions converged, none of them present today:

- **Laboratory validation** of the procedure on the equipment in use, with real samples.
- **Admission of the procedure into the execution rules.** The current kits do not provide for sample pooling in their instructions for use [10, 11], and the most recent procurement requires an automated methodology with operator interventions restricted to those provided for by the manufacturer [17]. The procedure would need to enter the instructions for use or be admitted by the contracting party.
- **Payment that reflects the number of reactions.** In current practice, payment is made once per sample with a released result, regardless of the number of repetitions [16, 17]. In this model, fewer reactions do not reduce the amount paid, and the per-result payment model creates no economic incentive for those who perform the tests to adopt the technique. The reduction would only reach the contracting party through a specific price for samples tested in pools or through price competition in a procurement that admits the procedure.
- **Eligible tests and logistics:** share of routine tests from people already suppressed, sufficient plasma volume and adapted laboratory workflows.

Long-term contracts reduce the opportunities to revisit these conditions. Until they converge, the possible gain is operational: equipment and bench time freed up in the laboratories.

V. DISCUSSION

A. Main findings

In this simulation, pools of 4 with retest only of quantified pools (rule Q20) saved 39% of reactions per collection and 50%

Table VIII. Longitudinal stage, main scenario, 17,301 episodes common to both arms: fewer reactions, confirmation within 60 days and within 240 days, and median days until confirmation. In bold, the main strategy.

Strategy	Fewer reactions	Confirmed within 60 days	Within 240 days	Median until confirmation
Individual	0%	96.9%	98.5%	42 days
Pool of 4, rule Q20	**50.5%**	**96.9%**	**98.7%**	**47 days**
Pool of 3, rule Q20	45.1%	97.1%	98.6%	47 days
Pool of 4, rule D	23.6%	97.0%	98.5%	47 days

Table IX. Confirmation within 60 days by rebound band, individual testing and pool of 4 with rule Q20, with residual RNA of 25%: mean and difference (pool minus individual, in points) with empirical interval across 40 replicates, in the main scenario and in the robustness scenario.

Rebound (cp/mL)	Episodes	Main: individual and pool	Main: difference	Robustness: individual and pool	Robustness: difference
200 to 249	686	44.7% and 43.7%	-1.0 (-6.7 to 4.2)	36.4% and 33.4%	-3.1 (-8.4 to 1.0)
250 to 299	569	78.7% and 78.7%	0.0 (-3.5 to 4.8)	59.3% and 56.1%	-3.2 (-8.1 to 0.7)
300 to 499	1,633	98.1% and 98.1%	0.0 (-0.8 to 0.9)	88.8% and 87.7%	-1.0 (-2.9 to 0.9)
500 to 999	2,182	100% and 100%	0.0 (0.0 to 0.0)	99.9% and 99.9%	0.0 (-0.2 to 0.2)
1,000 or more	12,231	100% and 100%	0.0 (0.0 to 0.0)	100% and 100%	0.0 (0.0 to 0.0)

during follow-up of suppressed people in the main scenario, and 26% to 42% per collection and 35% to 53% during follow-up across all the residual RNA scenarios tested. Confirmation of failure within 60 days stayed close to that of individual testing overall, with a median of 5 more days, which comes from the retest delay fixed in the model. Just above the cutoff, confirmation is low in both arms and the pool loses a little more: in the main scenario, 1 point between 200 and 249 cp/mL, with a wide interval; with greater imprecision, about 3 points between 200 and 299.

The finding that changed over the versions is the most important: the savings depend on the residual RNA below 50, which Brazilian public data do not inform. The rule that retests any detection (D), presented in version 1 as the main one, saves from 16% down to nothing when there is residual RNA. Rule Q20 keeps saving in all scenarios, but the magnitude depends on where this RNA is concentrated. Pre-concentration, designed to eliminate dilution, showed no advantage: with rule D, it loses the savings; with a trigger equivalent to that of Q20, it ties with the ordinary pool before the extra cost of concentrating.

B. Promising aspects and limits, side by side

This note does not use resampling of observed data or a registered success criterion; therefore, it does not calculate the chance of exceeding a reference. The promising aspects and the limits are these:

- **Promising:** rule Q20 kept savings in all the scenarios tested, from 26% to 42% per collection, with a magnitude sensitive to the distribution of loads below 50.
- **Promising:** difference of 0.3 point or less in confirmation within 60 days, across all episodes, in all combinations of risk, transient viremia and residual RNA.
- **Promising:** a rule executable with the output the kit already delivers to the reporting systems.
- **Limit:** just above the cutoff, confirmation within 60 days has wide intervals and a probable loss of about 3 points when the assay is less precise (Table IX).
- **Limit:** the intervals describe only the simulation's variation, not the uncertainty of the parameters.
- **Limit:** the share and shape of residual RNA in routine tests in Brazil are not known; the main scenario comes from studies in other countries.
- **Limit:** the reduction in reactions ranges from 36% to 56% depending on transient viremia and rebound risk, which have not been measured in routine tests of suppressed people in Brazil.
- **Limit:** the weight of the reaction in the price and the eligible fraction are not public, and the conversion into cost is illustrative.
- **Limit:** the technique is off-label use of the kit.
- **Limit:** the main configuration was chosen after comparing alternatives and after reviews, which requires confirmation with a registered protocol.

C. Comparison with the literature

Earlier studies reported from one third to more than half fewer reactions [3–6], and the implementation in Cameroon reached 0.55 assays per result with pools of 3 [8]. The 26% to 53% in this simulation fall largely within that range. Many earlier studies prioritized thresholds higher than the Brazilian one, although the evaluation by Smith and colleagues included the 250 cp/mL cutoff [3]. With the 200 cutoff and the current kits, a retest rule anchored in the limit of quantification performed

Table X. Main results in the three assay scenarios, pool of 4 with rule Q20 and residual RNA of 25%: fewer reactions per collection and during follow-up, and confirmation within 60 days (individual and pool).

Scenario	Per collection	During follow-up	Within 60 days
cobas 5800 (main)	39.4%	50.5%	96.9% and 96.9%
cobas 4800	39.5%	50.7%	97.1% and 97.2%
Robustness (greater imprecision)	39.2%	49.8%	95.1% and 94.9%

Table XI. Longitudinal sensitivity, main scenario, pool of 4 with rule Q20: annual rebound risk, share of collections with transient viremia, fewer reactions and confirmation within 60 days (individual and pool).

Rebound	Transient viremia	Fewer reactions	Within 60 days
3%	4%	56%	97.0% and 96.8%
3%	8%	47%	96.8% and 97.1%
6%	4%	51%	96.9% and 96.9%
6%	8%	42%	96.9% and 96.9%
10%	4%	44%	96.9% and 96.9%
10%	8%	36%	96.8% and 97.0%

better than quantitative deconvolution, which missed a large part of the loads just above the cutoff, and better than a rule anchored in simple detection, which is sensitive to the residual RNA that more sensitive kits have come to reveal [21]. Preconcentration has an operational precedent in blood banks [2] and in research with volumes of up to 30 mL [18], but, for this use, it only tied with the ordinary pool when given a trigger equivalent to that of Q20; with rule D, it revealed precisely the residual RNA that should not trigger a retest.

D. Strengths

- Assay parameters taken from the instructions for use of the two network kits, with the detection model checked against them.
- Retest rules restricted to those the kit allows to be executed.
- Explicit sensitivity analysis of the premise that weighed most on the savings (residual RNA below 50).
- Detection broken down by subrange and discordances in both directions.
- A longitudinal outcome with the same episodes in both arms, which counts missed episodes in the denominator.
- Explicit separation between reduction in reactions and reduction in cost, with the conditions for the latter.
- Results contrary to the hypotheses reported, including the correction of version 1.
- Code and numbers reproducible with fixed seeds.

E. Weaknesses of the method

- **Denominator.** The distribution in Table II includes everyone on treatment for six months or more. The technique would apply to routine tests of adults already suppressed, whose distribution was not found. With less low-level viremia, the reduction in reactions tends to be larger; with more, smaller.
- **Residual RNA.** The share was taken from studies in other countries, with different kits and definitions, and the shape is assumed; keeping the share at 25%, concentrating residual RNA close to 50 reduced the savings of Q20 from 39% to 26% per collection. Residual RNA was drawn again at each collection, with no persistence per person; if it is always concentrated in the same people, randomly assembled pools do not change the average result, but pools assembled by history could improve the savings.
- **Shape within ranges.** The uniform draw on a log scale within each range is an assumption; detection between 200 and 249 depends on it.
- **Imprecision at very low loads.** There are no data below 50 cp/mL on the 5800 or below 77 on the 4800; the value used is assumed. Rule Q20 is more sensitive to this than rule D, because it decides near 20 cp/mL in the pool; the robustness scenario shows the effect.
- **Dynamics.** Rebounds are sustained and at a constant level. Transient rebounds above 200, intermittent adherence failures and regimen changes are not included.
- **Laboratory.** Cross-contamination between samples, pipetting error, handling cost, bench time beyond the 5 days and adaptation of the reporting systems are not included. All of them tend to reduce the real gain.

Table XII. Gross scenario with 100% eligibility: estimated annual cost reduction (R × w × V), with V of about R$ 74 million, for the reduction in reactions per collection (39%) and during follow-up of suppressed people (50%), depending on the weight of the reaction in the price (w). Without deducting additional costs of the pooled workflow.

Reaction weight (w)	With 39% fewer reactions	With 50% fewer reactions
30%	11.8% (R$ 8.8 million)	15.2% (R$ 11.2 million)
50%	19.7% (R$ 14.6 million)	25.2% (R$ 18.7 million)
60%	23.6% (R$ 17.5 million)	30.3% (R$ 22.4 million)

- **Cost.** The cost structure of the price per test and the eligible fraction are not public; both were assumed.
- **Choice after looking.** The main configuration emerged from exploration and reviews; confirmation requires a registered protocol.

F. What the result allows us to conclude

It allows us to conclude that, with the public parameters available, a simple strategy of pools of 4 with retest of quantified pools is compatible with detecting failure at the 200 cp/mL cutoff and with the confirmation time frame, saving 26% to 53% of reactions depending on the residual RNA below 50, with more uncertainty in confirmation just above the cutoff. It also allows us to conclude that rules anchored in simple detection are not suitable for this use with the current kits, because they depend on the absence of residual RNA, and that pre-concentration showed no advantage. It does not allow us to claim performance in a real laboratory, nor to claim that there will be a cost reduction: that depends on the eligible fraction, on validation, on admission of the procedure and on payment linked to the number of reactions, conditions absent from current practice.

G. Next steps

Each path below only becomes work with a new protocol:

- obtain, from Brazilian routine data, the share of "detectado, inferior a 20" and "não detectado" results among suppressed people, and the persistence of these results per person;
- obtain the viral load distribution in routine tests of people who were already suppressed and the proportion of eligible tests;
- obtain imprecision data below 50 cp/mL;
- evaluate, with the trigger fixed before the run, triggers between 20 and 40 cp/mL in the pool, which may protect the savings against residual RNA close to 50 at the cost of more losses just above the cutoff;
- look for public estimates of the cost structure of viral load tests, to replace the illustrative reaction weight;
- include transient rebounds, plasma volume per sample and bench failures;
- carry out the systematic literature search, including validation studies of pooling on automated platforms;
- register the protocol before a confirmatory run.

VI. CONCLUSION

In this simulation with Brazilian public parameters and residual RNA in 25% of collections below 50 copies/mL, pools of 4 samples with retest only of quantified pools (20 cp/mL or more) used 39.7% fewer reactions per collection and 50.4% fewer during follow-up of suppressed adults. The difference in confirmation of failure within 60 days was close to zero (0.0 point; empirical interval -0.3 to 0.2), the median until confirmation was 47 days, against 42, and 2 of 1,106 loads between 200 and 249 were not retested. In all the residual RNA scenarios tested, this rule kept savings, from 26% to 42% per collection and from 35% to 53% during follow-up; the magnitude depends above all on how much of the residual RNA lies close to 50 cp/mL. Just above the cutoff, between 200 and 299 cp/mL, confirmation within 60 days has more uncertainty and falls by about 3 points when the assay is less precise. The rule that retests any detection, the main one in version 1, loses almost all its savings with residual RNA, and pre-concentration showed no advantage. The conversion into cost is conditional: with 100% eligibility, it would be 8% to 30% of the annual reference value, depending on the residual RNA and on the weight of the reaction in the price; with 60% eligibility, 5% to 18%; and it would only occur with laboratory validation, admission of the procedure and payment linked to the number of reactions. The note records the technique and these calculations for future studies.

VII. LIMITATIONS

- The research steps (literature search, reading of documents, code, simulation, analysis and writing) were carried out with artificial intelligence and did not undergo independent human validation. The results may contain errors or gaps because of this; version 1 contained a premise that was only corrected after publication.
- The design weaknesses listed in the Discussion also count as limitations.

- The note has no protocol registered before the analyses, and the configurations were chosen during exploration.
- The instructions for use validate the test for EDTA plasma (and, in the case of the 4800, also for the plasma separation card), without providing for sample pooling. Pooled testing would be off-label use.
- The distribution in Table II was read from a chart, by enlarged visual inspection.
- The annual reference value uses the contracted quantity, not the quantity performed.

VIII. GLOSSARY

- **Cohort:** group of people followed over time; here, simulated.
- **Confirmed virological failure:** two consecutive results of 200 cp/mL or more, with at least 4 weeks between collections.
- **Discordance:** a sample that one strategy classifies as 200 or more and the other does not.
- **Eligibility:** share of the test volume that could be tested in pools; here, routine tests of people already suppressed.
- **Empirical interval across replicates:** range of results obtained by repeating the simulation with different random draws.
- **Imprecision:** variation between repeated measurements of the same sample; here, standard deviation in log10.
- **Limit of detection:** lowest concentration detected in 95% of reactions.
- **Limit of quantification:** lowest concentration for which the kit reports a numerical value; below it, it reports only "detectado".
- **Lognormal:** distribution in which the logarithm of the value follows the normal curve; used for measurement errors proportional to the value.
- **Off-label use:** use of a diagnostic kit in a way not provided for in the instructions approved by the manufacturer.
- **Pooled testing:** testing a mixture of several samples and retesting individually only when the mixture shows a signal.
- **Pre-concentration:** gathering more plasma from each sample and concentrating the virus before the reaction, to offset dilution.
- **Quantitative deconvolution:** testing the samples of a positive pool one by one, subtracting the value of each from the estimated total, until the remainder falls below a threshold.
- **Reaction:** each time the instrument processes a tube, whether of a single sample or of a pool.
- **Reaction weight in the price:** share of the price per test that corresponds to the cost of the reaction (reagents and run), excluding equipment, collection, transport and services.
- **Residual RNA:** HIV RNA at a very low level, below 50 cp/mL, in people on effective treatment; the kit may report it as "detectado, inferior a 20".
- **Rule D:** retest individually the samples of every pool in which the kit detects virus.
- **Rule Q20:** retest only the samples of pools quantified at 20 cp/mL or more.
- **Seed:** number that fixes the sequence of random draws in a simulation, so that it can be repeated with the same result.
- **Transient viremia:** detectable, low viral load in a single collection, without treatment failure.
- **Viral load:** amount of HIV RNA per milliliter of plasma.

IX. REPORTING GUIDELINES

The note follows, insofar as it applies to simulation studies, the STRESS guideline [19] and, for the cost estimate, CHEERS 2022 [20]. The main items are in the sections: objectives and questions (I), sources and parameters (II), model logic, inputs, outcomes and variation (III), results with intervals, sensitivity analyses and economic potential (IV), limitations (V and VII) and reproducibility (Data and code availability).

DISCLAIMER

This is a working note in health informatics, produced with extensive use of artificial intelligence, and not a clinical study. The research steps carried out with AI (data collection, literature screening, analysis and code) have not undergone independent human validation, and the results may contain errors or gaps as a consequence; the author reviews the final text before publication. It has not been reviewed by a licensed health professional. It does not constitute advice, diagnosis, prognosis or treatment recommendation for any person. Hypotheses are hypotheses, not confirmed findings. The source data have limitations described in the text. Any use in health care requires specific review by licensed professionals, according to the intended use. Valid for the date and version indicated.

ARTIFICIAL INTELLIGENCE USE STATEMENT

Tools: Claude Opus 5.5 (Anthropic), in sessions on Nov 8 and 9, 2026 (versions 1, 1.1 and 1.2). Steps with AI: formulation of the hypotheses, literature search, queries to the

National Public Procurement Portal, reading of the public documents and of the kits' instructions for use, writing of all the code, running of the simulations, writing and translation into English and Spanish. Checking: automatic only: the numbers in the text were copied from the scripts' outputs, and the published scripts were run again to check that they reproduce those outputs; the detection model was compared with the tables in the instructions for use; excerpts from documents were checked by reading the PDFs. Reviews by another AI tool pointed out problems that were corrected before version 1 and, after it, the premise of zero load below 50 cp/mL, corrected in version 1.1; a second review pointed out the dependence on the range of residual RNA, the loss just above the cutoff and pre-concentration tested only with rule D, addressed in version 1.2; the calculations of both reviews were reproduced with the code published before the corrections. There was no independent human validation of the steps carried out by AI, and the results may contain errors or gaps because of this; the author reviews the final text. Tool limitations: it may make mistakes in code, in reading figures and in reading literature without the error showing up in the automatic checks. Record: code and outputs versioned in the program's public repository (github.com/willianwistuba/mais-saude-ai), in codigo_src/msa02_*.py and dados/msa02/; there was no protocol registered before the analyses. Conflicts and funding: none.

DATA AND CODE AVAILABILITY

All data used are public, and the scripts that produce every number are available at https://mais-saude-ai.vercel.app/en/code/, with instructions to reproduce the results.

ETHICS

This note uses only public, aggregated data that do not identify individuals, which waives ethics committee review under Brazilian National Health Council Resolution 510/2016 and CONEP Circular Letter 17/2022. Counts below five are suppressed at municipal level, and no re-identification is attempted.

COMPETING INTERESTS AND FUNDING

The author declares no competing interests. This work received no funding, and the project accepts no resources from manufacturers of technologies it assesses or comments on. This text has not been peer reviewed, reviewed by a health professional or subjected to independent human validation of the research steps carried out with AI (the author reviews only the final text), and should be read as a working note. Corrections create a new version, and the previous one remains accessible.

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